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Foundation or Formula? A Simulation-Based Comparison of Google's TimesFM-3 and Classical Time-Series Models

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Abstract

Google released TimesFM-3, a 330-million-parameter time-series foundation model, in August 2026. Whether such a model should displace ARIMA or exponential smoothing in applied work cannot be settled on public benchmarks, because only about 6% of the datasets used to evaluate published foundation models were absent from every model's pre-training corpus. We therefore generate the data. Under a known data-generating process the foundation model provably has not seen the series, and the classical model is correctly specified by construction, so the comparison is clean in both directions at once. Across 7 200 series from nine processes, four series lengths and a pre-registered protocol, neither family dominates: TimesFM-3 wins 16 of 36 cells and the classical methods 20, but those 20 are divided among five different winners while TimesFM-3 is a single untuned model, and it wins 113 of 132 significant pairwise comparisons. Only AutoARIMA holds its own. What TimesFM-3 buys is robustness rather than peak accuracy: it is never more than 1.31× worse than the best method in any cell, whereas every classical method is somewhere at least 2.2× worse. On strongly seasonal data with four or more observed cycles AutoARIMA is 21–24% better. On intermittent demand the metrics disagree, and adding six purpose-built Croston-family baselines sharpens rather than settles it: TimesFM-3 beats every one of them on MASE at every length and loses to them on RMSSE at every length, because absolute- and squared-error losses reward different functionals of a distribution that is 70% zeros. The decisive property proves to be not whether a classical model is correctly *specified* but whether it is *estimable*: given two seasonal cycles, AutoETS lost to the seasonal naive method by a factor of 2.7 on data it had itself generated. An analysis of 1 000 M4 Monthly series reproduces the pattern. Two findings cut across the comparison: TimesFM-3 is 103 times faster per series than AutoARIMA, and its weights are licensed for non-commercial use only, which removes it from production work entirely.

Keywords: time series forecasting, foundation models, TimesFM, ARIMA, exponential smoothing, simulation study, forecast evaluation.

1. Introduction

On 31 August 2026 Google Research released TimesFM-3, the third generation of its time-series foundation model (Sen, Zhou, Das, and Kong 2026). It is a decoder-only transformer with 330 million parameters, pre-trained on more than a trillion time points, and it is the first model in the family trained natively for *multivariate* forecasting: it consumes several related series together through alternating causal-temporal and full-variate attention layers, accepts past and past-future covariates, and emits an entire forecast horizon in a single non-autoregressive forward pass. Its authors report that it takes the top average rank among pre-trained foundation models on three public benchmarks, for both point and probabilistic accuracy.

For a practising forecaster the interesting question is not whether such a model tops a leaderboard. It is narrower and more useful: *given the series actually in front of me, should I use it, or should I use ARIMA?* A 330-million-parameter network and a three-parameter state-space model are not merely two points on an accuracy scale. They differ in what they cost to run, in whether their parameters mean anything, in whether their intervals can be justified from theory, and—as Section 5 discusses—in whether they may lawfully be used at all for a given purpose.

The published evidence is not well suited to answering that question, for a reason that has become difficult to ignore. Meyer, Kaltenpoth, Zalipski, and Müller (2025) traced the data lineage of 22 time-series foundation models and found that of the 401 datasets used to evaluate them, only about 6% had never appeared in *any* model's pre-training or fine-tuning corpus. They further showed that even a 0.1% overlap between pre-training and test data can move mean absolute percentage error by 8 to 29 percentage points, and that leakage need not be direct: a model trained on temporally overlapping, causally related series inherits an advantage on a target series it has never seen. A benchmark result obtained on a public dataset is therefore a measurement of unknown provenance. Compounding this, foundation-model papers have repeatedly been criticised for comparing against de-ensembled or under-tuned classical baselines rather than the combinations that win forecasting competitions (Makridakis, Spiliotis, and Assimakopoulos 2020).

This paper takes the one route that closes both objections at once: **it generates its own data**. Under a known data-generating process (DGP), two things become true simultaneously that are never true on a public benchmark. First, the foundation model provably has not seen the series, because the series did not exist until the experiment ran. Second—and this is the part that gives the comparison its edge—the classical model can be made *correctly specified by construction*. When the data really are ARIMA(1,1,1) with drift, AutoARIMA is not an approximation of the truth; it is the truth, up to parameter estimation.

One distinction must be drawn carefully here, because the paper's central claim depends on it. What the design excludes is contamination at the level of the *realisation*: these specific series did not exist before the experiment ran, so no pre-training corpus can contain them, and the leakage documented by Meyer *et al.* (2025) is impossible by construction. What the design does *not* exclude is familiarity at the level of the *family*. TimesFM-3 is trained on more than a trillion time points drawn from real-world *and synthetic* corpora (Sen *et al.* 2026), and synthetic pre-training data for time-series models is commonly generated from exactly the parametric processes D1–D7 use. The model may therefore arrive already fluent in ARMA-like and seasonal structure without ever having seen these draws.

That is not a flaw in the design; it is the boundary of what the design can claim, and it cuts in a specific direction. Family-level familiarity would tend to *help* TimesFM-3 on D1–D5, the processes where a classical model is correctly specified. Since those are precisely the cells where the classical methods win, the effect works against this paper's more favourable findings rather than for them. Where TimesFM-3 wins—intermittency, breaks—it wins on processes least likely to be over-represented in a synthetic ARMA corpus. Read strictly, then, the results below bound how much a foundation model gives up on classical home ground from *below*: with a less ARMA-fluent model the gap would be wider, not narrower.

That turns a vague question into a sharp one. Instead of asking which method is better in general, we can ask:

- how much accuracy does zero-shot TimesFM-3 give up on the classical model's *home ground*, where the parametric form is exactly right; and
- how much does it gain where no classical form is right—structural breaks, nonlinear saturation, intermittent demand, changing volatility?

We answer both across nine DGPs, four series lengths from 24 to 200 observations, three forecast horizons and 7 200 simulated series, comparing TimesFM-3 with seasonal naive, Theta, AutoETS, AutoARIMA and an equal-weight combination of the last three. The protocol, including the hypotheses, was pre-registered before any forecast was produced, and every departure from it is recorded.

We make no claim to a new estimator, test or theory. This is a comparative simulation study, and it is framed as one. Its contribution is evidential: a clean measurement of a trade-off that practitioners currently have to guess at, together with an explicit statement of the conditions under which each answer holds.

1.1. Related work and this paper's position

The foundation-model literature for time series is young and moving quickly. TimesFM (Das, Kong, Sen, and Zhou 2024), Chronos (Ansari, Stella, Türkmen, Zhang, Mercado, Shen, Shchur, Rangapuram, Arango, Kapoor, Zschiegner, Maddix, Wang, Mahoney, Torkkola, Wilson, Bohlke-Schneider, and Wang 2024) and Moirai (Woo, Liu, Kumar, Xiong, Savarese, and Sahoo 2024) established that a single pre-trained network can forecast unseen series without task-specific training, and GIFT-Eval (Aksu, Woo, Liu, Liu, Liu, Savarese, Xiong, and Sahoo 2024) supplied a common benchmark. The critical literature has grown alongside it: Meyer *et al.* (2025) on contamination, and Adler, Chang, Draxler, Abdi, and Smyth (2026) on calibration, who evaluated five foundation models against two baselines on six univariate datasets and found that calibration error grows with forecast length.

Our study is complementary to Adler *et al.* (2026) rather than a repeat of it, in three respects. Their evidence comes from public datasets and therefore inherits the contamination problem; ours is generated. They summarise reliability with calibration *metrics*; we pair interval reliability with pre-registered *hypothesis tests* on accuracy differences, with multiplicity control. And their study predates TimesFM-3, the first natively multivariate member of the family.

The classical side of the comparison is deliberately strong. AutoARIMA and AutoETS follow the Hyndman–Khandakar automatic selection procedures (Hyndman and Khandakar 2008; Hyndman, Koehler, Snyder, and Grose 2002), Theta is the method that won M3 (Assimakopoulos and Nikolopoulos 2000), and the equal-weight combination is included because combinations, not individual models, are the standard against which competition entries are judged (Makridakis *et al.* 2020). Evaluation follows the scale-free conventions of Hyndman and Koehler (2006) and the M5 competition (Makridakis, Spiliotis, and Assimakopoulos 2022a; Makridakis, Spiliotis, Assimakopoulos, Chen, Gaba, Tsetlin, and Winkler 2022b), with the pitfalls catalogued by Hewamalage, Ackermann, and Bergmeir (2023) treated as constraints rather than suggestions.

2. Design of the simulation study

2.1. Data-generating processes

Nine processes were chosen to span the range from "a classical model is exactly right" to "no classical model is right". Table 1 lists them. Each series is generated at length $n + H$ with $H = 12$; the first n observations form the context supplied to every method, and the final 12 are held out as the truth. One realisation of each process is shown in Figure 1.

Table 1: The nine data-generating processes. The final column states which classical family is correctly specified by construction; for D6–D8 none is, and for D9 the mean is correctly specified while the conditional variance is not.

ID	Data-generating process	Parameters	Correctly specified by
D1	AR(1)	$\phi = 0.7$	ARIMA
D2	ARMA(1,1)	$\phi = 0.6$, $\theta = 0.4$	ARIMA
D3	ARIMA(1,1,1) with drift	$\phi = 0.5$, $\theta = 0.3$, drift 0.3	ARIMA
D4	SARIMA$(1,0,0)(1,1,0)_{12}$	$\phi = 0.6$, $\Phi = 0.5$	ARIMA
D5	ETS(A,A,A)	$\alpha = .3$, $\beta = .01$, $\gamma = .2$	ETS
D6	Local level with a break	break at $0.7n$, jump 10σ	neither
D7	Logistic growth	$K = 180$, $r = 0.08$, floor 20	neither
D8	Intermittent demand	$P = 0.3$, size $1 + \text{Pois}(4)$	neither
D9	AR(1) with GARCH(1,1) errors	$\phi = .6$, $a = .1$, $b = .85$	ARIMA in mean

Figure 1: One realisation of each data-generating process (n = 96; held-out horizon in colour)

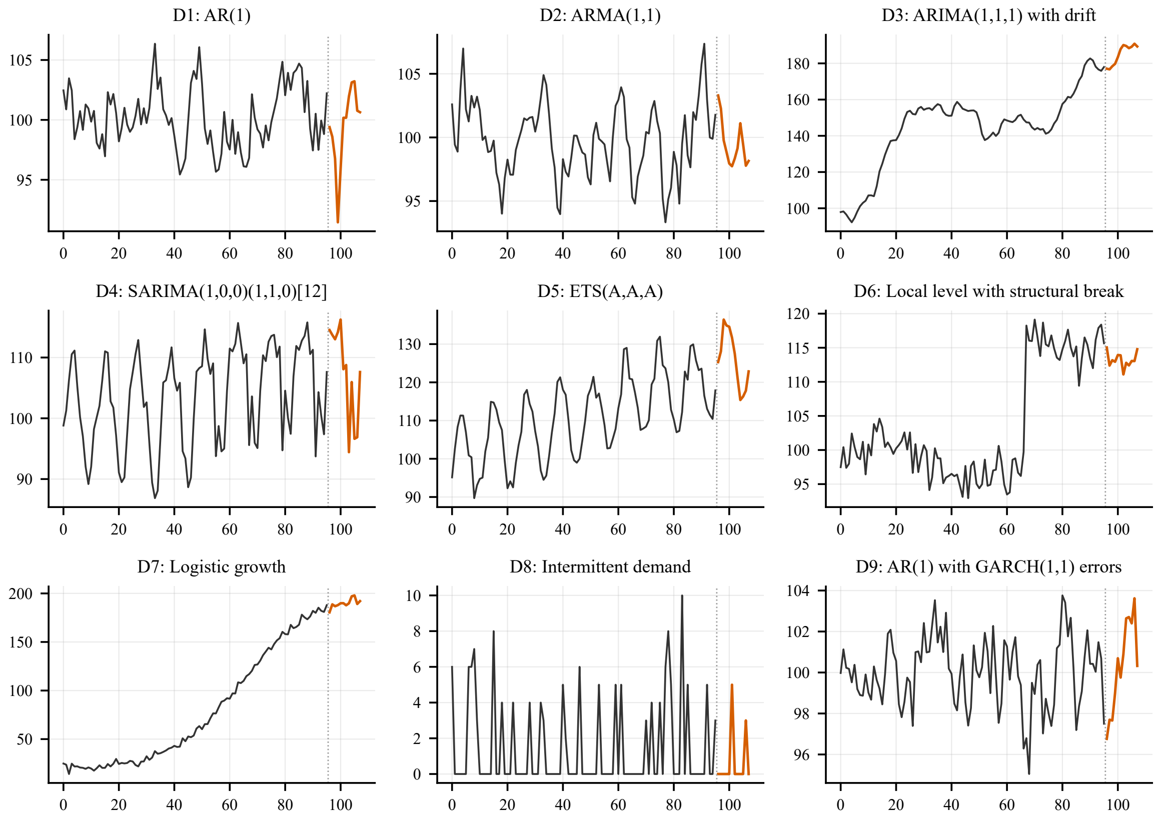

Figure 1: One realisation of each data-generating process at $n = 96$. The context supplied to every method is in grey and the held-out horizon in colour, separated by the dotted line.

Three choices in Table 1 deserve comment, because each was made to keep the comparison honest rather than to favour an outcome.

The structural break in D6 is placed inside the *context*, at $0.7n$, not inside the forecast window. A break in the held-out window is unforecastable by any method, so a design that placed it there would measure only which method is the most conservative. Placing it in the context instead poses a question every method can answer: having seen the level shift, do you forecast from the new level or revert towards the old one?

The inflection of the logistic curve in D7 sits at $0.6(n + H)$, so the held-out window falls in the saturating arm. Any method that fits a linear drift should overshoot here; that is the behaviour the process is designed to expose.

Series lengths are $n \in \{24, 48, 96, 200\}$. The shortest is deliberately punishing: at $n = 24$ with a seasonal period of 12, a seasonal model has only two complete cycles from which to estimate its seasonal structure. As Section 3 shows, this separates two things that are usually conflated—whether a model is correctly *specified*, and whether it is *estimable* from the data available.

Two hundred replications are generated per (DGP, length) cell, giving $9 \times 4 \times 200 = 7\,200$ series. Each series is seeded deterministically from its own identifiers, so it is reproducible independently of core count or execution order.

2.2. Methods compared

No method receives hand-tuning; each classical model uses its own automatic order and parameter selection, which is the comparison a practitioner would actually face. TimesFM-3 is used zero-shot, with no fine-tuning, through the released `google/timesfm-3.0-pytorch` weights (Google Research 2026); the classical models come from **StatsForecast** (Garza, Canseco, Challú, and Olivares 2022).

The equal-weight combination of Theta, AutoETS and AutoARIMA is not an optional extra. The most frequently levelled criticism of foundation-model evaluations is that they are compared against individual, de-ensembled classical methods, when the classical state of the art is a combination. Excluding it would build the paper's conclusion into its design.

How the combination's *predictive distribution* is formed should also be stated, since one recommendation rests on it. Its point forecast is the equal-weight mean of the three members, but its quantiles are obtained by averaging the members' quantile functions decile by decile (vincentization) rather than by mixing their distributions. The two differ: vincentization produces a narrower predictive distribution than a mixture when the members disagree about location, so the combination's intervals here are sharper—and, as Section 4 shows, better calibrated on M4—than a mixture would have given. Any comparison of the combination's intervals against TimesFM-3's is a comparison against this particular construction.

One asymmetry in the setup should be stated plainly, because it favours the classical side. Every classical method is given the *true* seasonal period as an input—$m = 12$ on D4 and D5, $m = 1$ elsewhere—whereas TimesFM-3 receives only the raw context, with no frequency label and no indication that the series is seasonal at all. The classical methods thus begin with a piece of oracle knowledge the foundation model must infer. This reflects ordinary practice, since an analyst usually does know whether data are monthly, and the same m is used for the MASE denominator so the metric is unaffected. But it means the seasonal results (D4, D5) are, if anything, generous to the classical methods, and a like-for-like comparison in which the period had to be estimated would narrow their advantage there.

2.3. Evaluation

Point accuracy is measured primarily by the mean absolute scaled error (MASE; Hyndman and Koehler 2006), with the root mean squared scaled error (RMSSE) and sMAPE as secondary measures. All three are scaled by the in-sample seasonal-naive error computed on the context, so they are comparable across processes of different magnitude.

MAPE is reported only in Appendix A. It is undefined when an actual value is zero, it is asymmetric between over- and under-forecasts, and on the intermittent process D8 it is simply unusable—a point the results make concrete rather than assert (Hyndman and Koehler 2006; Kolassa 2016).

Probabilistic accuracy is measured by the mean pinball loss over the nine deciles $q = 0.1, \ldots, 0.9$, scaled by the same denominator, together with the empirical coverage and mean width of the nominal 60% and 80% intervals. The choice of 60% and 80%, rather than the more familiar 90% or 95%, is forced by the comparison itself: TimesFM-3 emits only the nine deciles, so a

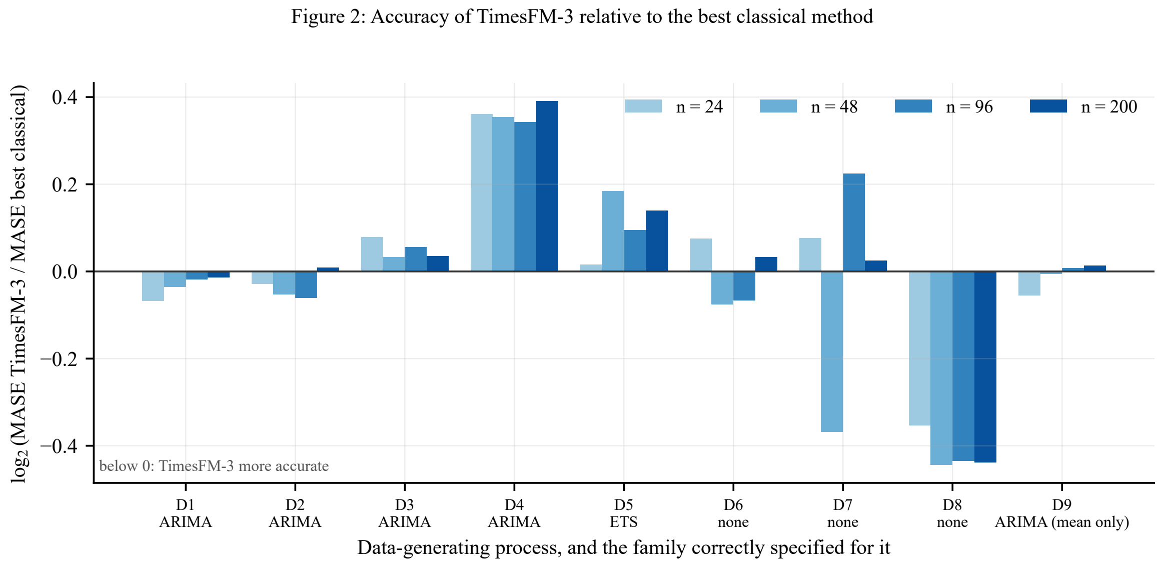

Figure 2: Accuracy of TimesFM-3 relative to the best classical method in each cell, on a $\log_2$ scale. Bars below zero mark cells where TimesFM-3 is more accurate. The label under each process names the family correctly specified for it by construction.

90% interval cannot be formed from its output. Comparing a classical 90% interval against a foundation-model interval constructed by extrapolation would not be a like-for-like comparison, so the widest interval both families can express, $[q_{0.1}, q_{0.9}]$, is used as the widest interval reported. Classical quantiles are obtained from prediction intervals at levels 20, 40, 60 and 80, which map exactly onto the same deciles.

Differences in accuracy are tested, not merely tabulated. Within each (DGP, length, horizon) cell, TimesFM-3 is compared with each classical method by a paired Wilcoxon signed-rank test (Wilcoxon 1945) on per-replication MASE differences, accompanied by the Hodges–Lehmann estimate of the median difference (Hodges and Lehmann 1963) and the median MASE ratio, so that a difference which is statistically detectable but practically negligible is visible as such. Multiplicity across cells is controlled by the Benjamini–Hochberg procedure at a false discovery rate of 0.05 (Benjamini and Hochberg 1995).

The Diebold–Mariano test (Diebold and Mariano 1995), which is the usual instrument in this literature, is *not* used anywhere in this paper, and it is worth saying why since its absence is conspicuous. DM is built for an autocorrelated sequence of loss differentials generated by a single series over time. In the simulation tier the replications within a cell are independent draws from the DGP, so a paired rank test across replications is the appropriate tool and DM would misstate the dependence structure. The pre-registered protocol did reserve DM for the real-data tier, where rolling origins do generate such a sequence—but the realised design yields only three origins per series, giving a per-series DM statistic two degrees of freedom and no useful power. Applying it instead across the 1 000 series would be worse still, since those are indexed by series identifier rather than by time, and a serial-correlation correction over an arbitrary ordering is not a test at all. DM is therefore dropped in both tiers rather than reported with a caveat, and the departure is recorded.

3. Results

The design yields 7 200 series, and with six methods and three horizon slices, 129 600 per-series evaluations. Table 3 reports mean MASE over the full horizon $h = 1, \ldots, 12$; Figure 2 shows the same information as the accuracy of TimesFM-3 relative to the best classical method in each cell.

Table 2: Mean MASE over $h = 1, \ldots, 12$ across 200 replications per cell. Lower is better; the best method in each row is in bold. The parenthesised family after each DGP label is the one correctly specified for it by construction.

DGP	n	Seas. naive	Theta	AutoETS	AutoARIMA	Combin.	**TimesFM-3**
D1 (ARIMA)	24	1.646	1.682	1.568	1.485	1.538	**1.417**
	48	1.625	1.586	1.570	1.415	1.478	**1.381**
	96	1.626	1.572	1.564	1.324	1.435	**1.307**
	200	1.615	1.576	1.576	1.266	1.417	**1.254**
D2 (ARIMA)	24	1.905	2.003	1.886	1.708	1.788	**1.675**
	48	1.838	1.860	1.834	1.575	1.688	**1.518**
	96	1.810	1.812	1.811	1.517	1.643	**1.455**
	200	1.922	1.919	1.921	**1.374**	1.654	1.383
D3 (ARIMA)	24	**4.411**	4.483	4.892	5.275	4.659	4.656
	48	4.252	**4.179**	4.547	4.679	4.313	4.274
	96	4.125	4.135	3.938	3.940	**3.871**	4.023
	200	4.512	4.427	4.275	**4.195**	4.199	4.297
D4 (ARIMA)	24	**1.029**	4.146	3.602	2.284	3.084	1.321
	48	1.021	1.944	1.720	**1.014**	1.395	1.295
	96	1.005	2.910	2.407	**0.886**	1.839	1.124
	200	1.053	4.155	1.085	**0.886**	1.700	1.161
D5 (ETS)	24	**1.139**	3.154	3.025	1.574	2.474	1.152
	48	1.108	0.956	0.741	**0.717**	0.731	0.815
	96	1.076	0.757	0.761	0.732	**0.713**	0.761
	200	1.040	0.759	0.681	**0.626**	0.645	0.689
D6 (none)	24	0.801	1.327	**0.766**	0.802	0.837	0.807
	48	0.947	1.115	0.869	0.925	0.924	**0.825**
	96	1.006	0.963	0.905	0.920	0.916	**0.865**
	200	0.934	0.840	**0.825**	0.833	0.827	0.844
D7 (none)	24	5.417	3.179	1.272	**1.083**	1.680	1.142
	48	4.243	2.262	1.602	1.076	0.970	**0.751**
	96	1.655	**0.821**	1.305	1.329	1.097	0.959
	200	**0.917**	1.301	0.966	0.994	0.954	0.933
D8 (none)	24	1.019	0.999	1.047	0.996	1.010	**0.780**
	48	1.066	0.984	0.988	0.962	0.974	**0.708**
	96	0.932	0.923	0.926	0.924	0.923	**0.683**
	200	1.005	0.943	0.939	0.939	0.940	**0.694**
D9 (ARIMA (mean only))	24	1.547	1.518	1.454	1.372	1.404	**1.321**
	48	1.588	1.486	1.476	1.244	1.353	**1.239**
	96	1.430	1.342	1.333	**1.098**	1.213	1.104
	200	1.577	1.503	1.504	**1.190**	1.341	1.200

Table 3: Mean MASE over $h = 1, \ldots, 12$ across 200 replications per cell. Lower is better; the best method in each row is in bold. The parenthesised family after each DGP label is the one correctly specified for it by construction.

Table 4: Significant pairwise comparisons at $h = 1, \ldots, 12$, Benjamini–Hochberg corrected within each opponent family. TimesFM-3 dominates every classical method except AutoARIMA, against which it is close to even.

Opponent	Significant comparisons	TimesFM-3 wins	Opponent wins
AutoARIMA	20	11	9
Seasonal naive	26	22	4
AutoETS	27	25	2
Theta	29	28	1
Combination	30	27	3

3.1. The headline is a split decision, not a rout

Counting the winner in each of the 36 (DGP, length) cells gives TimesFM-3 16 and the classical methods 20—AutoARIMA 10, seasonal naive 4, and two each for Theta, AutoETS and the combination. Tested against the *best* classical method in each cell, 18 of the 36 differences survive Benjamini–Hochberg correction: 7 favour TimesFM-3 and 11 favour the classical method. The remaining 18 cells are ties.

Those "versus best classical" p-values should be read as descriptive rather than as valid tests, and we flag this rather than let the reader assume otherwise. The comparator is selected as the minimum-MASE classical method *using the same 200 replications* on which the test is then computed, so the comparison is post-selection inference: selecting a minimum over five candidates and then testing against it biases the comparison in the classical methods' favour, and the quoted adjusted p-value carries no correction for that selection. The correctly adjusted pairwise comparisons in Table 4, where each opponent is fixed in advance, do not suffer from this and are what the paper's inferential claims rest on. The cell-winner count is reported because it is the quantity a practitioner intuitively asks for, not because it is the more rigorous statistic.

That is the cautious reading. The more useful reading emerges from the 180 pairwise comparisons, of which 132 are significant, 113 favouring TimesFM-3 and 19 the classical method. Broken down by opponent, the pattern is stark:

Table 4 carries the study's central message. **AutoARIMA is the only classical method that holds its own**; against everything else—including the combination of all three—TimesFM-3 wins the large majority of significant comparisons. A practitioner who reliably selects AutoARIMA loses nothing by staying classical. A practitioner who does not, or cannot, is materially better off with the foundation model.

3.2. Effect sizes, and a sensitivity check that matters

The evaluation protocol promised a Hodges–Lehmann estimate and a paired t-test alongside every rank test, and both are reported here rather than left in the results files.

The Hodges–Lehmann estimates confirm that the significant differences are also small. Median shifts in MASE, across the 36 cells, are -0.010 against AutoARIMA, -0.091 against the combination, -0.133 against seasonal naive, -0.183 against AutoETS and -0.227 against Theta (negative favours TimesFM-3). Against AutoARIMA the typical difference is about one hundredth of a MASE unit—detectable in places, but not a difference a forecaster would notice.

The t-test sensitivity check agrees with the Wilcoxon test on significance in 163 of 180 comparisons (90.6%). The 17 disagreements are not noise and one pattern among them is substantive: on the intermittent process D8, the paired t-test declares TimesFM-3 significantly better than seasonal naive at every length ($p < 0.0001$) while the signed-rank test finds no significant difference ($p = 0.11$ to 0.92). The two tests are answering different questions. The

Table 5: Each method's mean MASE divided by the best mean MASE in the same cell, summarised over the 36 cells. A value of 1.00 means the method was the best in that cell. TimesFM-3 is never worse than 1.31× the best method anywhere, and in the median cell it is within 0.8% of it; every classical method is at least 2.2× worse somewhere.

Method	Median	Mean	Worst	Worst cell
SeasonalNaive	1.200	1.470	5.65	D7 $n = 48$
Theta	1.251	1.614	4.69	D4 $n = 200$
AutoETS	1.194	1.366	3.50	D4 $n = 24$
AutoARIMA	1.032	1.137	2.22	D4 $n = 24$
Combination	1.108	1.273	3.00	D4 $n = 24$
TimesFM-3	**1.008**	**1.054**	**1.31**	D4 $n = 200$

mean difference is large because a minority of series produce very large errors for seasonal naive; the median difference is near zero because on most individual series the two methods are close. On intermittent data, where the error distribution is heavily skewed, this gap is expected, and it is a further reason—beyond the metric disagreement above—to treat the D8 result as conditional rather than decisive.

3.3. Specification is not estimability

The pre-registered hypothesis H1 held that classical models would win on their home ground—where they are correctly specified by construction—with the gap largest at short series lengths. **H1 is refuted in that form, and the reason is instructive.**

On D1 (AR(1)) and D2 (ARMA(1,1)), TimesFM-3 is at least as accurate as the correctly specified AutoARIMA at every length except $n = 200$ in D2, and none of those differences except that one is significant. Being correctly specified bought AutoARIMA nothing detectable on short and moderate series, because the parameters still have to be estimated from the data at hand, whereas TimesFM-3 brings a prior learned from more than a trillion time points.

The seasonal cells make the point unmistakable. At $n = 24$ with a period of 12—two complete cycles—AutoETS scores a mean MASE of **3.03 on D5, data that an ETS process generated**, while the seasonal naive method scores 1.14 and TimesFM-3 scores 1.15. The correctly specified model lost to a one-line rule by a factor of nearly three. The same happens on D4, where Theta records 4.15 at $n = 200$ against AutoARIMA's 0.89.

So the operative distinction is not whether a classical model is correctly *specified*, but whether it is *estimable* from the series in front of you. Where both hold, the classical model wins. Where specification is right but estimation is starved, automatic classical methods do not merely lose—they fail loudly, and TimesFM-3 degrades gracefully instead.

Table 5 quantifies that asymmetry. Expressing each method's mean MASE as a multiple of the best mean MASE in the same cell, **TimesFM-3 is never worse than** 1.31× **the cell's best method anywhere in the design**, and in the median cell it is within 0.8% of it. Every classical method is at least 2.2× worse somewhere: AutoARIMA 2.22× and AutoETS 3.50× (both on D4 at $n = 24$), Theta 4.69×, seasonal naive 5.65×.

We use the ratio to the cell's best method rather than the absolute worst MASE because the absolute measure is uninformative here: D3 is difficult for every method, so absolute worst cases are dominated by that process rather than by any method's fragility. On the absolute measure Theta would appear the most robust of all, which inspection of D4 shows it plainly is not.

3.4. Two decisive regimes, one of which depends on the metric

Away from the ties, two regimes separate at every length. One of them is not as clean as a

Table 6: The four scale-free metrics disagree on the intermittent process D8. MASE and the scaled pinball loss favour TimesFM-3 at every length; RMSSE and sMAPE favour the classical methods at every length. The best value in each row is in bold.

Metric	n	Seas. naive	Theta	AutoETS	AutoARIMA	Combin.	**TimesFM-3**
MASE	24	1.019	0.999	1.047	0.996	1.010	**0.780**
	48	1.066	0.984	0.988	0.962	0.974	**0.708**
	96	0.932	0.923	0.926	0.924	0.923	**0.683**
	200	1.005	0.943	0.939	0.939	0.940	**0.694**
RMSSE	24	0.940	0.791	0.792	**0.765**	0.775	0.806
	48	0.937	0.746	0.726	**0.718**	0.724	0.774
	96	0.875	0.707	**0.697**	0.698	0.697	0.775
	200	0.923	0.718	**0.706**	0.706	0.706	0.797
sMAPE	24	**86.696**	172.741	169.988	165.908	170.489	191.774
	48	**84.690**	172.522	170.746	170.622	171.026	196.183
	96	**84.171**	170.801	169.574	170.065	169.949	197.610
	200	**85.990**	170.003	169.189	169.315	169.236	198.556
Pinball	24	0.634	0.395	0.402	0.380	0.388	**0.325**
	48	0.653	0.381	0.368	0.360	0.365	**0.307**
	96	0.614	0.348	0.341	0.341	0.341	**0.288**
	200	0.628	0.354	0.344	0.345	0.345	**0.291**

single metric makes it appear, and we report that rather than the favourable number alone.

Strong seasonality with enough data to estimate it (D4): use AutoARIMA. At $n = 48, 96, 200$ the best classical method is AutoARIMA and it is better than TimesFM-3 by 21.8%, 21.1% and 23.7% respectively, all with adjusted $p < 0.0001$. Once at least four seasonal cycles are available, AutoARIMA recovers the seasonal structure and TimesFM-3 cannot match it. D5 tells the same story more mildly for $n \geq 48$.

The recommendation reverses at $n = 24$, and the reversal is instructive. With only two seasonal cycles the best classical method is not AutoARIMA at all but the seasonal naive rule (1.029), and **TimesFM-3 beats AutoARIMA there by 42%** (1.321 against 2.284). "Use AutoARIMA for seasonal data" is therefore sound advice only once the seasonal model is estimable; below that threshold it is poor advice, which is the same lesson as Section 3.3.

Intermittent demand (D8): the metrics disagree, and the disagreement is the finding. On MASE, TimesFM-3 reduces error against the best classical method by 21.7%, 26.5%, 26.0% and 26.2% at the four lengths, with adjusted $p < 0.0001$ throughout; on the scaled pinball loss it is likewise best at every length. But on RMSSE it *loses* at every length, and on sMAPE it is the worst method of the six by a wide margin (Table 6).

This is not a contradiction in the data; it is a known property of intermittent series, and it has a mechanism. Roughly 70% of D8's observations are zero. Absolute-error metrics such as MASE are minimised by the conditional *median* of the predictive distribution, which for this process is zero or near it, while squared-error metrics such as RMSSE are minimised by the conditional *mean*, which is strictly positive. A method that forecasts close to the median therefore wins MASE and loses RMSSE, and one that forecasts the mean does the reverse. This is exactly the trap Kolassa (2016) documents for count-valued retail demand, and it means no single number settles the intermittent case.

What can be said honestly is narrower than our pre-registered hypothesis H2 anticipated: on intermittent demand TimesFM-3 produces the better *median-like* point forecast and the better full predictive distribution as scored by a proper scoring rule, while the classical methods produce the better mean-like forecast. Which one a practitioner wants depends on the loss function their decision actually faces—an inventory cost that is linear in units favours the former, one that is quadratic favours the latter. We resisted the temptation to headline the

Table 7: Intermittent-demand baselines on D8, added post hoc. TimesFM-3 beats every purpose-built method on MASE at every length (24 of 24 comparisons significant after Benjamini–Hochberg correction), and loses to them on RMSSE at every length. The best value in each row is in bold.

Metric	n	Croston	Croston-opt	Croston-SBA	ADIDA	IMAPA	TSB	AutoARIMA	**TimesFM-3**
MASE	24	1.042	1.024	1.025	0.972	0.972	0.971	0.996	**0.780**
	48	0.990	0.992	0.974	0.947	0.954	0.966	0.962	**0.708**
	96	0.935	0.948	0.921	0.926	0.923	0.916	0.924	**0.683**
	200	0.947	0.948	0.934	0.941	0.942	0.948	0.939	**0.694**
RMSSE	24	0.769	0.758	0.763	**0.753**	0.757	0.764	0.765	0.806
	48	0.722	0.726	0.718	**0.717**	0.722	0.733	0.718	0.774
	96	0.696	0.702	**0.694**	0.700	0.701	0.716	0.698	0.775
	200	0.709	0.710	0.707	0.709	0.712	0.730	**0.706**	0.797

MASE result alone.

3.5. Adding the right tool for the job: Croston-family baselines

The pre-registered comparison set contains no method designed for intermittent demand. Croston's method (Croston 1972) and its descendants exist precisely for this regime, and their absence would leave the D8 result showing only that TimesFM-3 beats *general-purpose* classical methods on data none of them was built for. We therefore added them and re-ran the comparison. This analysis is post hoc and is reported as such.

Six intermittent-demand methods were added: Croston's classic and optimised variants, the Syntetos–Boylan approximation (Croston-SBA), ADIDA, IMAPA and TSB. Table 7 gives the result, and it does not soften the earlier finding—it sharpens it in both directions.

On MASE, **TimesFM-3 beats every one of the six purpose-built methods at every series length**: all 24 comparisons are significant after Benjamini–Hochberg correction, with median ratios between 0.70 and 0.82. The best intermittent baseline at any length is TSB at $n = 96$ (0.916), against 0.683 for TimesFM-3. So the objection that TimesFM-3 was merely beating the wrong tools does not survive: it also beats the right ones, and by a wider margin than it beat the general-purpose set.

On RMSSE the reversal not only persists but becomes cleaner. ADIDA and Croston-SBA record the lowest RMSSE at every length (0.694–0.769) against TimesFM-3's 0.774–0.806. This is the sharpest available confirmation of the mechanism proposed above. Croston-family methods are constructed to estimate the mean demand *rate*, which is exactly the functional that squared-error loss rewards; TimesFM-3's forecast sits nearer the conditional median, which is what absolute-error loss rewards. Two families optimised for different functionals of the same predictive distribution each win under the metric aligned with their own target, and neither result is an artefact.

The practical conclusion is correspondingly conditional, and it is now supported by the correct baselines rather than asserted against inappropriate ones. If the inventory or staffing decision downstream has a cost roughly linear in units, TimesFM-3 is the better choice on this process by a wide and significant margin. If that cost is closer to quadratic, a Croston-family method is, and TimesFM-3 should not be used.

3.6. Behaviour across the forecast horizon

Figure 3 averages mean MASE within each regime at $h = 1$, $h = 1, \dots, 6$ and $h = 1, \dots, 12$. Every method degrades with horizon, as expected, and the ordering is stable across horizons—nothing reverses as the forecast lengthens.

The figure needs one caution, because read carelessly it appears to contradict Table 3. TimesFM-3 has the lowest averaged MASE in *both* panels, including the panel where clas-

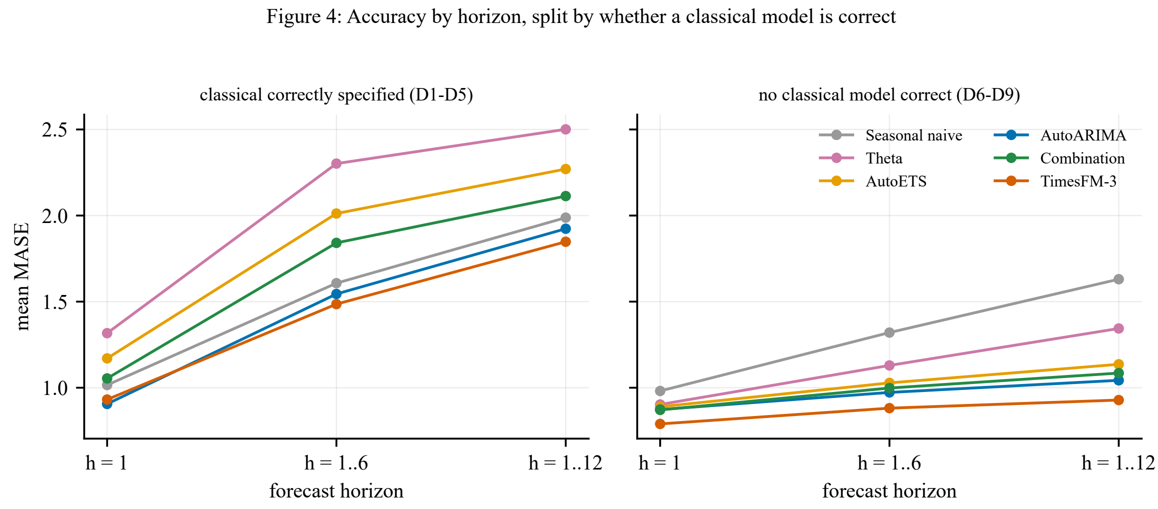

Figure 3: Mean MASE by forecast horizon, averaged within each regime. Left: the five processes for which a classical family is correctly specified. Right: the four for which none is. The averages are dominated by the cells in which an automatic classical method fails badly, which is why TimesFM-3 leads both panels even though the classical methods win more individual cells.

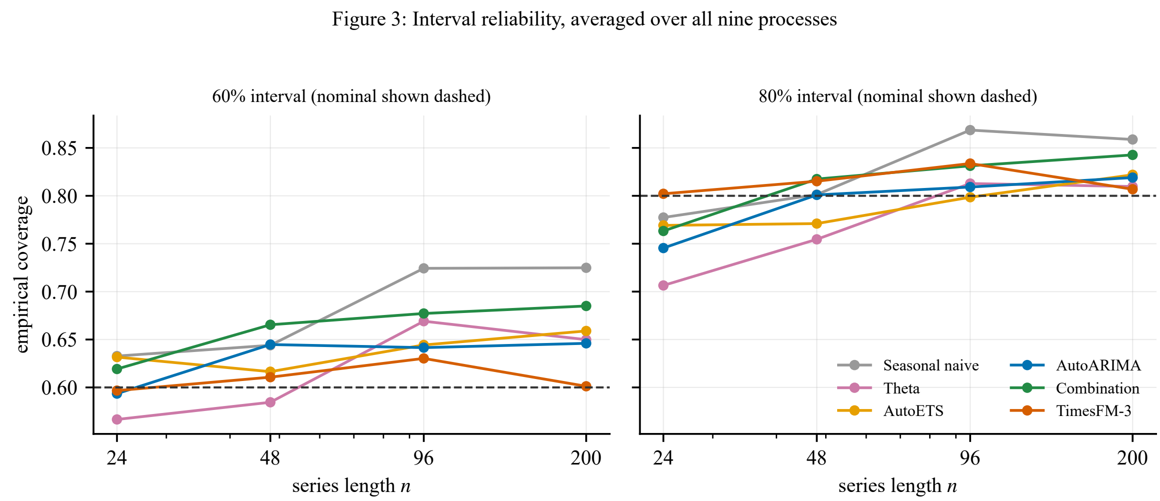

Figure 4: Empirical coverage of the nominal 60% and 80% intervals, averaged over all nine processes, against series length. The nominal level is dashed. TimesFM-3's 80% coverage stays within 0.802–0.834 across every length; the classical methods under-cover at $n = 24$ and over-cover from $n = 96$.

sical models are correctly specified, yet the cell-by-cell count in Section 3 gives the classical methods more wins. Both are true and they are not in tension: averaging over a regime mixes the cells where a classical method is best with the cells where it collapses, and a single catastrophic cell (Theta at 4.16 on D4, AutoETS at 3.03 on D5) moves a mean far more than several narrow wins do. The average rewards consistency; the cell count rewards peak performance. That gap between the two is the robustness result of Table 5 seen from another angle, not a separate finding.

3.7. Interval reliability

Figure 4 reports empirical coverage against nominal, averaged over the nine processes. On that averaged view TimesFM-3's 80% coverage stays between 0.802 and 0.834 across series lengths, closer to nominal than any other method, while seasonal naive drifts from 0.777 at $n = 24$ to 0.869 at $n = 96$.

That averaged view is, however, flattering to every method, and we state the honest figure alongside it. Averaging coverage across processes lets over-coverage on one process cancel under-coverage on another, so the deviation of an average understates the typical error. The appropriate statistic is the mean of the per-cell absolute deviations, and it is roughly three times larger: 0.047 for TimesFM-3, 0.066 for AutoARIMA, 0.089 for the combination, 0.101 for AutoETS, 0.127 for Theta and 0.144 for seasonal naive. The ranking is unchanged and TimesFM-3 remains the best calibrated, but the level is not close to nominal: **its per-cell 80% coverage ranges from 0.583 to 0.921**. No method in this study delivers reliable 80% intervals across all nine processes, and it would be wrong to read Figure 4 as saying otherwise.

The structural-break process D6 is the sharpest illustration, and it shows why coverage alone is a misleading summary. Every classical method covers between 0.966 and 0.999 there against a nominal 0.80. That is not reliability; it is a symptom. The break inflates the estimated innovation variance, so the intervals become enormous—scaled widths of 4.8 to 10.4, against 3.0 to 5.1 for TimesFM-3. The classical intervals contain the truth almost always because they contain almost everything. TimesFM-3 covers between 0.828 and 0.889, still above nominal, using intervals roughly half as wide.

3.8. Robustness: was the combination a straw man?

TimesFM-3 beats the pre-registered equal-weight combination in 27 of 30 significant comparisons, which invites the objection that the combination was weak—it contains Theta, which fails badly in several cells, and a mean is not robust to one broken member. We therefore repeated the comparison against a *median* combination of the same three methods. The objection does not hold: the median combination is slightly worse than the mean combination (mean MASE 1.712 against 1.656, with TimesFM-3 at 1.439), and TimesFM-3 still wins 25 of 29 significant comparisons. This analysis is post hoc and is reported as such.

3.9. What MAPE would have told you

MAPE is undefined for 12 906 of the 129 600 evaluations—9.96%, and every one of them on the intermittent process D8. A study that had used MAPE as its headline metric would have had to drop the single regime in which TimesFM-3 wins most decisively, or silently substitute a convention for the zeros. This is the concrete form of a long-standing warning (Hyndman and Koehler 2006; Kolassa 2016), and it is why MAPE appears here only in Appendix A.

4. Real data: M4 Monthly

The simulation tier buys freedom from contamination at the cost of realism. This section pays some of it back, with the caveat stated up front rather than buried: **M4 is contamination-suspect**. It is a widely used public benchmark and is very likely inside TimesFM-3's pre-training corpus (Meyer *et al.* 2025). Any advantage TimesFM-3 shows here may therefore be memory rather than skill, which is exactly why the simulation tier carries the paper's primary claim. This tier answers a different and narrower question: which of the simulated regimes do real monthly series resemble?

4.1. Design

We draw a seeded random sample of 1 000 series from the 32 581 M4 Monthly series long enough for the design, and evaluate each at three rolling origins spaced 12 months apart with M4's own monthly horizon, $h = 18$. Three origins per series would give 3 000 windows; the realised count is 2 932, because a series shorter than 138 observations cannot support the

Table 8: M4 Monthly, 1 000 series at up to three rolling origins each (2 932 windows in all), $h = 18$. Paired Wilcoxon across series, Benjamini–Hochberg corrected. The pattern matches the simulation: TimesFM-3 beats every classical method except AutoARIMA, against which it ties—and here the equal-weight combination also ties.

Comparison	Mean MASE	Median ratio	% series better	Adj. p	Verdict
TimesFM-3	0.914	—	—	—	
vs seasonal naive	1.280	0.699	80.1	< 0.0001	TimesFM-3
vs Theta	0.970	0.933	56.8	< 0.0001	TimesFM-3
vs AutoETS	0.951	0.966	54.1	0.0043	TimesFM-3
vs AutoARIMA	0.923	0.970	52.6	0.27	tie
vs combination	0.907	0.996	48.5	0.27	tie

third origin once 18 held-out points and a 120-observation minimum context are subtracted, and those 68 windows are dropped rather than shortened. The sampled series are substantial: median length 299 observations, upper quartile 324, maximum 1 808. This is deliberately not toy data.

The pre-registered protocol specified the Diebold–Mariano test here. With only three origins per series that statistic carries two degrees of freedom and has essentially no power, so the analysis is instead a paired Wilcoxon test across the 1 000 series—which are independent units, unlike the overlapping windows within a series. Diebold–Mariano is not reported at all; the reasons are given in Section 2.3 and the departure is recorded.

4.2. Results

Table 8 gives the outcome, and it reproduces the simulation's pattern closely. TimesFM-3 significantly outperforms seasonal naive (better on 80.1% of series), Theta (56.8%) and AutoETS (54.1%), and is *statistically indistinguishable* from both AutoARIMA (52.6%, adjusted $p = 0.27$) and the combination (48.5%, adjusted $p = 0.27$).

Two observations follow.

First, **the tiers agree**. M4 Monthly series are long (median 299 observations) and seasonal—precisely the regime in which the simulation found classical methods competitive or better, because there is ample data with which to estimate the right model. Real monthly data sits in the estimable regime, and the real-data result is what Section 3 predicts for it. The agreement is worth more than either tier alone, since the two have opposite weaknesses.

Second, **the probabilistic result splits**, and the two halves point opposite ways. TimesFM-3's interval *coverage* does not carry over: on simulated data it was the best calibrated, whereas here it is the worst but one, covering 0.765 against a nominal 0.80 while the combination achieves 0.801. Yet on the pre-registered proper scoring rule—the scaled pinball loss over the nine deciles—TimesFM-3 is **the best of the six methods** (0.363, against 0.365 for the combination and 0.374 for AutoARIMA), even though the combination edges it on mean MASE.

These are not in conflict; they measure different things. Coverage asks only whether the truth falls inside one interval, and is blind to how the remaining probability mass is arranged; the pinball loss scores the whole predictive distribution across all nine deciles. TimesFM-3's distribution is therefore better *overall* while its 80% interval is too narrow. A reader who cares about a calibrated interval should prefer the combination here; a reader who cares about the full predictive distribution should prefer TimesFM-3. Reporting only the coverage figure would have understated it, and reporting only the pinball loss would have overstated it. The fragility of foundation-model calibration reported by Adler *et al.* (2026) is visible in the coverage half of this result.

Table 9: Wall-clock cost per series, each method timed separately on 200 seasonal series of length 96 ($h = 12$). TimesFM-3 was run on an NVIDIA GTX 1660 Ti; its one-off model load of 3.0 s is excluded, as are the classical methods' import costs.

Method	Time per series (ms)	Relative to TimesFM-3
Seasonal naive	3.1	0.13×
Theta	9.0	0.39×
TimesFM-3 (GPU, batched)	**23.2**	1×
AutoETS	102.0	4.4×
AutoARIMA	**2382.7**	**102.8**×

5. Costs that do not appear in an accuracy table

Accuracy is one input to the decision and, for many deployments, not the binding one.

5.1. Computational cost, which runs the opposite way to expectation

It is natural to assume that a 330-million-parameter network is the expensive option and a three-parameter statistical model the cheap one. On this hardware that is emphatically false. Table 9 reports each method timed separately on the same 200 series.

AutoARIMA is **103 times slower per series than TimesFM-3**. The reason is structural rather than incidental: automatic ARIMA runs a stepwise search over model orders, fitting each candidate by maximum likelihood, separately for every series. TimesFM-3 performs one batched forward pass and amortises it across the batch. Where a practitioner's constraint is throughput over many series—which is exactly the setting in which Section 3 found the foundation model most attractive—the accuracy argument and the cost argument point the same way.

Memory was also pre-registered as an operational cost, and the honest report is that our measurement of it is weak. Peak Python heap allocation during fitting was about 1 MB for every classical method (0.6 MB for Theta, 1.4 MB for AutoARIMA), but that instrument counts only Python-level allocations and sees neither the compiled **numba** kernels the classical methods run in nor the GPU memory TimesFM-3 uses, so no comparable figure for TimesFM-3 can be given. What can be said without measurement is structural: the classical methods hold a handful of parameters per series, while TimesFM-3 requires its 330-million-parameter checkpoint resident throughout—roughly 1.3 GB—which is the binding memory constraint in any deployment and is invariant to how many series are being forecast.

Two qualifications keep this honest, and the second matters more than the headline ratio. TimesFM-3's figure depends on a GPU; without one the comparison narrows substantially. And the classical methods parallelise trivially across series, which recovers most of the difference: in the study's own main run, fitting all four classical methods over all 7 200 series took 7.3 minutes on 22 cores, against 3.2 minutes for TimesFM-3 on a single consumer GPU—a ratio of 2.3×, not 103×.

The per-series figure and the end-to-end figure answer different questions, and quoting only the first would overstate the case. The fair summary is that TimesFM-3 is dramatically cheaper than AutoARIMA per series on one core, and modestly cheaper than the whole classical suite once that suite is given a 22-core machine. What it is not, on any measure taken here, is the expensive option.

5.2. Licensing

The most consequential difference between the two families is not statistical. TimesFM-3's pre-trained weights are released under `timesfm-non-commercial-license-v1.0` and are restricted to non-commercial, non-production use (Google Research 2026). The source code remains Apache-2.0, and so were the weights of TimesFM-2.5, but the version-3 weights are not.

The practical consequence is absolute and prior to any accuracy consideration: **for a commercial forecasting application, TimesFM-3 as released is unavailable at any accuracy**. A 26% MASE reduction on intermittent demand is irrelevant to a retailer who may not lawfully deploy the model that achieves it. The available responses are to use TimesFM-2.5 under Apache-2.0, to negotiate separate terms, or to use a classical method. Every classical method compared here is free of such restrictions.

We note this not as criticism but because it is routinely omitted from model comparisons, and it inverts the ranking for an entire class of users.

5.3. Interpretability and diagnosis

An ARIMA fit reports its orders, its coefficients and their standard errors; an ETS fit reports its smoothing parameters and states. These support the ordinary diagnostic work of applied forecasting—inspecting residuals, testing for remaining autocorrelation, explaining to a stakeholder why the forecast turns where it does, and detecting when a model has stopped fitting. TimesFM-3 exposes none of this. When it forecasts poorly there is no parameter to inspect and no diagnostic to run; the available response is to notice the errors and switch methods. In regulated settings where a forecast must be justified rather than merely produced, this difference may outweigh the accuracy comparison entirely.

5.4. Prediction intervals

The classical intervals derive from a model's assumed error structure and can be reasoned about analytically. TimesFM-3's quantiles are learned outputs of a network head, with no accompanying theory. Section 3 shows the learned quantiles are in fact better calibrated on average—but empirically, on these processes, not by construction, and their behaviour on a process unlike anything in the pre-training corpus is unknown by definition.

6. When to use which

Table 10 states the study's operational conclusion. It is deliberately narrow: it applies to the regimes examined, and it says "either" wherever the evidence does not support a preference.

7. Discussion

7.1. What the evidence supports, and what it does not

Of the five pre-registered hypotheses, **none survives unqualified**: two are refuted outright, two are partially supported, and one is refuted in its second clause.

- **H1** (correctly specified classical models beat TimesFM-3 on their home ground, gap largest at short lengths) — **refuted as stated**. The gap at short lengths runs the other way. What survives is narrower: where a classical model is both correctly specified *and* estimable, it wins (D4 and D5 at $n \geq 48$).

- **H2** (TimesFM-3 wins wherever no classical form is correct) — **partially supported**. It holds on intermittent demand under MASE and pinball loss but not under RMSSE or sMAPE, mildly on structural breaks, and not on the logistic process, where the outcome depends on length.

Table 10: When to prefer which family, from the evidence in Section 3. Bold marks a preference supported by significant differences; the remaining rows rest on considerations outside the accuracy comparison.

Situation	Use	Evidence
Commercial or production deployment	Classical (or TimesFM-2.5)	Version-3 weights are licensed non-commercial; accuracy is not the binding constraint
Intermittent demand, and the decision loss is roughly linear in units	**TimesFM-3**, but see the caveat	Beats all six Croston-family methods on MASE at every length (24/24 significant), and the general-purpose set by 22–26%. But WORSE on RMSSE and sMAPE (D8)
Intermittent demand, and the decision loss is roughly quadratic	Classical	The metrics reverse: Croston-SBA and ADIDA win on RMSSE at every length (D8)
Strong seasonality, at least four full cycles observed	**AutoARIMA**	AutoARIMA 21–24% better; TimesFM-3 27–31% worse, all $p < 0.0001$ (D4). Reverses below four cycles
Short series with seasonality (two cycles or fewer)	Seasonal naive or TimesFM-3	Automatic classical methods fail badly here: AutoETS scores 3.03 on its own process (D5, $n = 24$)
Many heterogeneous series, no capacity to select and tune a model per series	**TimesFM-3**	Beats every classical method except AutoARIMA in the large majority of significant comparisons
One series, and you can identify and estimate the right model	Classical	AutoARIMA is even with TimesFM-3 (11–9) and is free, interpretable and diagnosable
Structural break in the recent past, intervals matter	**TimesFM-3**	Comparable coverage using intervals roughly half as wide (D6, simulation only)
Throughput matters and a GPU is available	**TimesFM-3**	103× faster per series than AutoARIMA; 2.3× faster than the whole classical suite on 22 cores
Well-calibrated intervals on real data are the priority	Classical combination	On M4 the combination covered 0.801 against nominal 0.80; TimesFM-3 under-covered at 0.765
The forecast must be explained or audited	Classical	TimesFM-3 exposes no parameters, residual diagnostics or interval theory

- **H3** (on D9 the families are indistinguishable in the mean, classical intervals under-cover) — **partially supported**. The families are indeed indistinguishable in the mean, but AutoARIMA is the only classical method that under-covers on D9; the others over-cover.

- **H4** (the combination beats every individual classical method and is the hardest baseline for TimesFM-3) — **refuted in both clauses**. AutoARIMA beats the combination on mean MASE, and AutoARIMA, not the combination, is the baseline TimesFM-3 finds hardest (11–9, against 27–3 for the combination).

- **H5** (both families under-cover at 80%) — **refuted**. On the process-averaged view TimesFM-3 sits close to nominal and the classical methods over-cover from $n = 96$; on the honest per-cell view every method's typical deviation is far larger than expected, but not systematically downward.

That every hypothesis failed in some respect is the strongest argument for having pre-registered them. Had the predictions been written after the results, this section would have reported five confirmations.

7.2. Interpreting the split decision

It would be easy to read 16 cell wins against 20 as "the classical methods still win". That reading is available but shallow, because the 20 are distributed across five different methods—AutoARIMA 10, seasonal naive 4, and two each for the others. The classical column of Table 3 is a column of *different* winners, and which one wins is not knowable in advance without the kind of model-selection work the foundation model is meant to remove. TimesFM-3, one model with no tuning, is within noise of that per-cell best in half the design and beaten decisively in only two regimes.

The honest summary is therefore neither "foundation models have won" nor "classical methods remain superior". It is that **TimesFM-3 buys robustness, not peak accuracy**, and Table 5 puts a number on the trade: it is within 0.8% of the cell-best method in the median cell and never more than 1.31× worse anywhere, while every classical method is somewhere at least 2.2× worse. It is rarely the best method and never a bad one, whereas the automatic classical methods are sometimes excellent and sometimes catastrophic—AutoETS at 3.03 on data generated by an ETS process is the cautionary example. For a forecaster with one well-understood series, that trade is unattractive. For a forecaster with ten thousand heterogeneous series and no capacity to diagnose each one, it is the whole argument.

7.3. Limitations

Five limitations bound these conclusions.

First, the simulation tier's strength is also its boundary. Series generated from nine parametric processes are not real series; they are clean, stationary where specified, and free of the measurement error, calendar effects, promotions and regime changes that characterise applied work. What the simulation buys—provable absence of contamination and known correct specification—it pays for in realism, which is why Section 4 exists.

Second, TimesFM-3 was evaluated zero-shot and univariate. Its headline novelty is native multivariate forecasting with covariates, and that capability is untested here because the simulated processes are univariate by construction. A study exercising cross-series information could reach different conclusions, and would be the natural sequel.

Third, no method received hand-tuning. That is deliberate and it is the comparison most practitioners face, but an expert who identified each process by eye and fitted the corresponding model by hand would beat every automatic method reported here, including on D4 and D5 at $n = 24$.

Fourth, the horizon is short ($H = 12$) and the seasonal period fixed at 12. Longer horizons and other frequencies may behave differently, and the calibration literature reports that foundation-model calibration degrades with horizon (Adler *et al.* 2026).

Fifth, the intermittent-demand analysis of Section 3 is post hoc: the Croston-family baselines were added after the main results were seen, in response to an identified gap, and are reported separately from the pre-registered comparison.

Sixth, this is a study of one foundation model. Chronos (Ansari *et al.* 2024) and Moirai (Woo *et al.* 2024) are not evaluated, so nothing here should be read as a claim about foundation models in general.

8. Conclusion

We compared TimesFM-3 with five classical alternatives on 7 200 series generated from nine known processes, under a pre-registered protocol, in a design where contamination of the foundation model is impossible by construction and the classical models are correctly specified by construction.

Neither family dominates. TimesFM-3 wins 16 of 36 cells and the classical methods 20, but the 20 are spread across five different winners while TimesFM-3 is a single untuned model. Against individual classical methods it wins 113 of 132 significant comparisons, and only AutoARIMA holds its own at 11–9. Measured against the best method in each cell, TimesFM-3 is never more than 1.31× worse anywhere in the design and is within 0.8% of the best in the median cell, while every classical method is somewhere at least 2.2× worse. On strongly seasonal data with at least four observed cycles AutoARIMA is 21–24% better. On intermittent demand the answer depends on the loss function: TimesFM-3 is about a quarter better on MASE and best on the scaled pinball loss, but worse on RMSSE and much worse on sMAPE, because absolute- and squared-error metrics reward different functionals of a predictive distribution that is 70% zeros.

A secondary analysis of 1 000 M4 Monthly series reproduces the pattern closely: TimesFM-3 beats seasonal naive, Theta and AutoETS significantly, and ties with AutoARIMA and the combination. Real monthly series are long and seasonal—the estimable regime—and the tiers agree on what happens there, which is worth more than either tier alone given their opposite weaknesses.

The most transferable finding is not about either family in particular. It is that the decisive property is not whether a classical model is correctly *specified* but whether it is *estimable*: on two seasonal cycles, AutoETS lost to a one-line rule by a factor of 2.7 on data it had itself generated. Foundation models are most valuable exactly where classical estimation is starved, and least valuable where it is comfortable.

Two findings outside the accuracy comparison deserve equal weight. TimesFM-3 is 103 times faster per series than AutoARIMA—the foundation model is the cheap option, not the expensive one, though parallelising the classical suite across 22 cores narrows the end-to-end gap to 2.3×. And its released weights may not be used commercially. For a large class of readers that second point settles the question before any of the above becomes relevant.

9. Computational details and reproducibility

All results were produced with Python 3.13 (Python Core Team 2025), using **statsforecast** 2.1.1 (Garza *et al.* 2022) for the classical methods and **timesfm** 3.0.1 with the `google/timesfm-3.0-pytorch` weights (Google Research 2026) on **PyTorch** 2.6 (Paszke, Gross, Massa, Lerer, Bradbury, Chanan, Killeen *et al.* 2019). The simulation, evaluation and figures rest on **NumPy** (Harris, Millman, van der Walt *et al.* 2020), **pandas** (McKinney 2010), **SciPy** (Virtanen, Gommers, Oliphant *et al.* 2020) and **Matplotlib** (Hunter 2007). Timings were taken on an NVIDIA GTX 1660 Ti with a 24-core CPU and 32 GB of memory; the loaded TimesFM-3 checkpoint reports 330 710 976 parameters.

Every simulated series is seeded deterministically from its own identifiers—process, length and replication—so the entire study regenerates exactly from the code, independently of core count or execution order. The analysis pipeline is split into cached phases, so each stage can be re-run alone and the results reproduced without repeating the whole computation.

Three points of research practice are worth stating explicitly, since they bear on how much weight the results carry. The protocol, including the five hypotheses, was frozen before any forecast was produced, and the frozen file was never edited afterwards; nine departures from it are recorded separately, each dated and reasoned, with the one post-hoc analysis marked as such. Every identifier in the bibliography was checked by resolving it, rather than written from memory. The check parses the bibliography file itself and resolves each DOI through `doi.org` content negotiation, which answers for both registration agencies—necessary because arXiv DOIs are issued by DataCite and are invisible to the Crossref API. Two traps motivated this. A title-based Crossref *search* returns the wrong record for several of the classical references, including the working-paper version of Diebold and Mariano (1995) and a later reprint of Hodges and Lehmann (1963); and an earlier checker that maintained its own list of DOIs in parallel with the bibliography silently drifted out of step with it. Works published at venues that mint no DOI—PMLR, TMLR, OpenReview and the Curran NeurIPS proceedings—are cited by their proceedings URL, with the preprint identifier given in a note. And the two hypotheses the data refuted are reported in Section 3 at the same prominence as those it supported.

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Declarations

Funding. No funding was received for this work. **Competing interests.** The author declares no competing interests. **Data and code availability.** The complete research compendium is archived on Zenodo at `10.5281/zenodo.22681072` (concept DOI, resolving to all versions): the pre-registered protocol, the deviation log, the entire pipeline, the per-series metrics for all 129 600 evaluations, the raw forecasts and quantiles, every figure and the manuscript source. All data in the simulation tier are generated by that code from documented seeds; the real-data tier uses the publicly available M4 competition data (Makridakis *et al.* 2020), which the code re-fetches, with the seeded 1 000-series sample included so the analysis reproduces exactly.

The raw forecasts are deposited deliberately and not only for completeness. This paper's intermittent-demand result reverses between absolute- and squared-error metrics, so a reader who wants to re-score the same forecasts under a different loss can do so without re-running the study. TimesFM-3's weights are not redistributed: they are released by Google under a non-commercial licence and are downloaded by the code from Hugging Face.

A. MAPE, and why it is not the headline metric

Table 11 reports mean MAPE by process, alongside the percentage of series on which it cannot be computed. The D8 row is the argument: every intermittent series contains a zero, so MAPE is undefined for all of them, and the process on which TimesFM-3 achieves its largest and most consistent advantage simply drops out of the comparison.

The rankings also move. On D4, MAPE places seasonal naive first (2.360) and TimesFM-3 second (2.801), whereas MASE places AutoARIMA first at every length from $n = 48$. MAPE's asymmetry between over- and under-forecasts is enough to reorder the methods on the same forecasts.

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Table 11: Mean MAPE over $h = 1, \ldots, 12$, with the percentage of series on which MAPE is undefined. On the intermittent process D8 every series contains a zero, so MAPE cannot be computed at all and the entire regime in which TimesFM-3 wins most decisively disappears from the comparison.

DGP	% undefined	Seas. naive	Theta	AutoETS	AutoARIMA	Combin.	**TimesFM-3**
D1	0%	2.785	2.737	2.681	2.336	2.501	**2.283**
D2	0%	3.291	3.341	3.277	2.696	2.971	**2.641**
D3	0%	9.509	9.278	9.875	10.078	9.393	**9.214**
D4	0%	**2.360**	7.471	4.986	2.862	4.538	2.801
D5	0%	2.971	3.583	3.270	2.436	2.916	**2.256**
D6	0%	2.011	2.379	1.835	1.902	1.925	**1.822**
D7	0%	7.634	4.631	2.979	2.538	2.767	**2.184**
D8	100%	—	—	—	—	—	—
D9	0%	1.861	1.769	1.741	1.470	1.599	**1.457**

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